MILITARY SPECIFICATION SHEET

MOBILE FACILITY: SHELTER, INSTRUMENTATION AND CONTROL, TURBO SHAFT ENGINE TEST SYSTEM

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

The complete requirements for procuring the Mobile Facility described herein shall consist of this document and the issue in effect of MIL-M-81957(AS).

REQUIREMENTS

Size - The internal dimensions of the mobile facility (hereinafter referred to as "The Unit") shall be as follows:

Length	232	(+1.0; -0) inches
Width	89.812	(+1.5; -0) inches
Height	84.312	(+2.5; -0) inches

Weight ratings -

Tare weight - Tare weight of the unit shall not exceed 4,400 pounds.

Maximum payload - The unit shall be designed and constructed to operate with a maximum payload of 15,600 pounds.

Design gross weight (R) - The design gross weight (tare weight plus maximum payload) on which the structural design and testing are based shall be 20,000 pounds except that 44,800 pounds shall apply for stacking requirements.

Design load factors - The design load factors shall be in accordance with (American National Standard Institute) ANSI MH 5.1, except that all factors for Air-rotary wing shall be deleted.

Personnel doors - The unit shall contain two personnel doors. One door shall be located at each end of the unit. Both doors shall be seventy-six inches high. One door shall be thirty-five inches wide, the other shall be twenty-eight inches wide (see Figure 1). The distance from the bottom of the doors to the floor shall be not greater than 3/16 of an inch.

Door hinges - The door hinges shall be located on the right hand side when facing the doors from the outside. The doors shall open to the outside.

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Safety glass - The doors shall contain a 12" x 12" x 1/4" piece of safety glass (see Figure 1).

Safety glass cover - The inside of each door shall be equipped with a hinged metal cover capable of closing off all light coming through the window. The hinge shall be on the bottom of the cover.

Observation window - An observation window shall be provided as shown in Figure 1. The observation window assembly shall consist of two panes of safety glass, one 5/16 inches thick and the other 1/4 inch thick. (The height and length as specified in Figure 1.) The two panes shall be separated by a 1/2 inch air space which will be acoustically sealed and demoisturized to prevent fogging. The 5/16 inch thick glass shall be laminated security glass in accordance with MIL-G-3787, Class I, Type I. The window assembly shall be mounted to permit expansion and contraction of the glass while maintaining a vapor tight seal between the glass and the frame.

Stuffing box - An opening for the stuffing box panel shall be provided in accordance with Figure 1 and Figure 4.

Air conditioning system interface - The openings for the air conditioning system shall be in accordance with Figure 1 and Figure 4.

Power panel - An opening for an electrical power panel shall be provided in accordance with Figure 1 and Figure 3.

Protective finish - The unit shall be given a protective finish in accordance with MIL-STD-808. This includes hardware such as handles, hinges, screws, etc., and the necessary touchup after mounting. The type of finish shall be as follows:

Exterior - F304DG - Type I

Ceiling - F304XL - Type II

Walls - F304WL - Type II

Floor - F304QL - Type II

Acoustical requirements - When subjected to the acoustical absorption test specified in ASTM C423, and being exposed to the noise levels specified in Table I, the sound pressure level inside the unit shall not exceed the levels specified in Table II.

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TABLE I

Octave Band Center Frequency	63	125	250	500	1000	2000	4000	8000
SPL in dB re 0.0002 dynes/cm^2	110	115	120	122	120	116	114	110

TABLE II

Octave Band Center Frequency	63	125	250	500	1000	2000	4000	8000
SPL in dB re 0.0002 dynes/cm^2	79	73	68	64	62	60	59	57

Panels (Construction) - Shelter panels (ends, roof, walls, floor and doors) shall be constructed with a double-skin metallic material bonded to a core material constructed to meet all requirements to this specification. See MIL-M-81957(AS).

Skids - Three (3) removable skids, three inches in height and three inches wide, shall be provided on the underside of the unit and located in accordance with Figure 5. The two outer skids shall each be equipped with four floating (+1/4 inch) stainless steel, 5/8-11 UNC, captivated nuts located as shown in Figure 5. The ends of the skids shall be welded closed. Each skid shall be of a length to permit being placed on the floor, inside the unit. (For transportation) The skids shall be capable of supporting the design gross weight of the unit and shall meet the interchangeability requirements of MIL-M-81957(AS). The increase in height of the unit (with the skids installed) shall be not greater than three inches.

Identification of product - In lieu of the identification specified in MIL-M-81957(AS) the unit shall be marked in accordance with Naval Air Systems Command Drawing 981AS309.

Location - The identification plate shall be located to the left of the air conditioner openings (see Figure 1) at a height of 60 inches from the bottom of the unit.

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QUALITY ASSURANCE

Skidbearing test - The unit, loaded to the design gross weight shall be balanced on a two inch diameter pipe located anywhere along the three skids. The skids shall experience no fractures or ruptures and no permanent deformation greater than 1/4 inch as a result of this test.

(Project No. 2330-N648)

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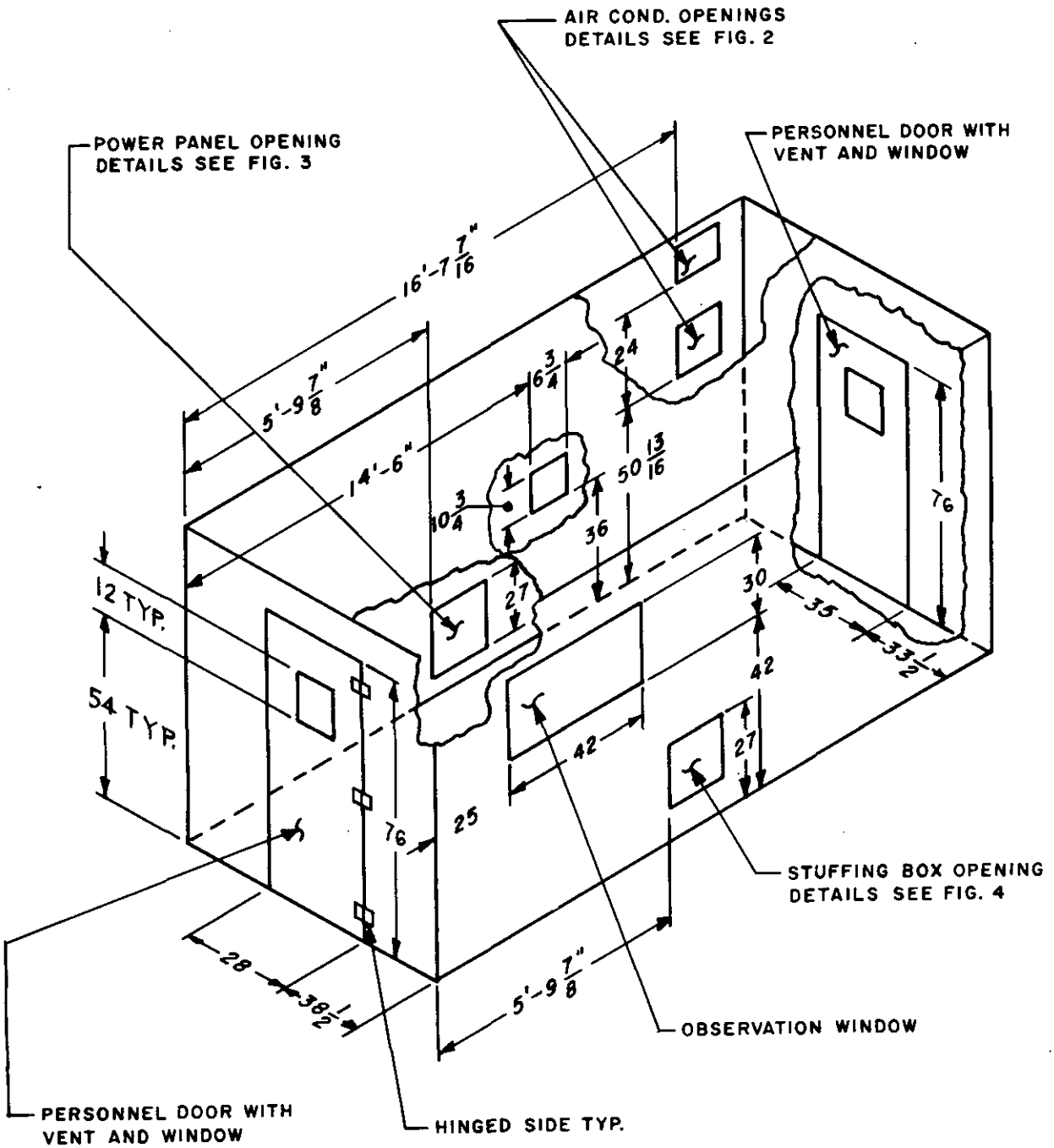
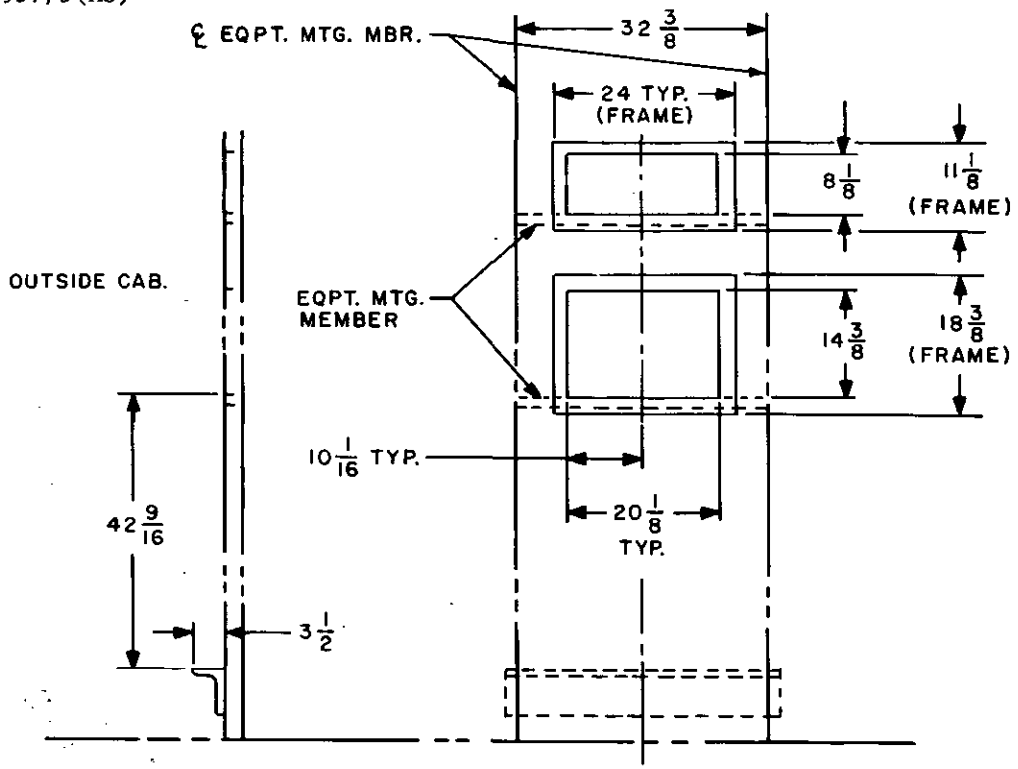


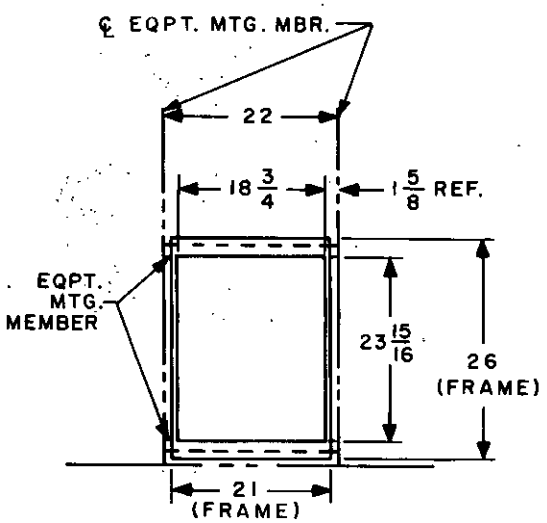
FIGURE 1. DIMENSIONS FROM INSIDE SURFACES

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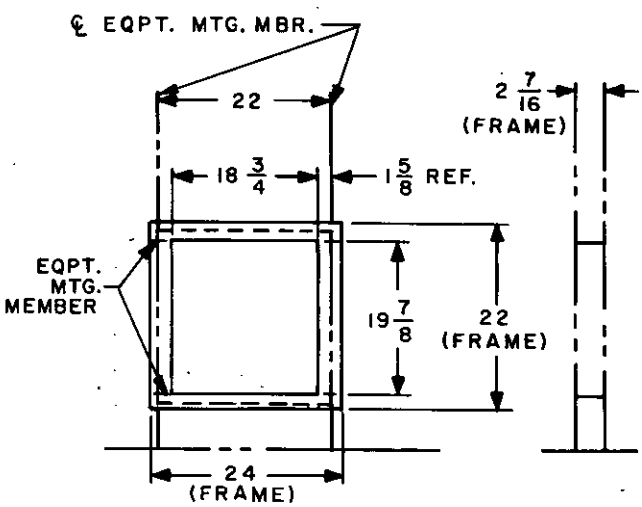
AIR COND. OPENING DETAILS (450 LBS)

FIGURE 2



POWER PANEL OPENING DETAILS (50 LBS)

FIGURE 3



STUFFING BOX OPENING DETAILS (50 LBS)

FIGURE 4

NOTE: ALL OPENINGS SHALL BE FRAMED AND SUPPORTED TO MAINTAIN STRUCTURAL INTEGRITY WHEN EQUIPMENT IS MOUNTED TO OPENING INTERFACE. APPROXIMATE WEIGHTS OF EQUIPMENT ARE SHOWN. RIVETS SHOULD BE FLUSH WITH INTERFACE SURFACES.

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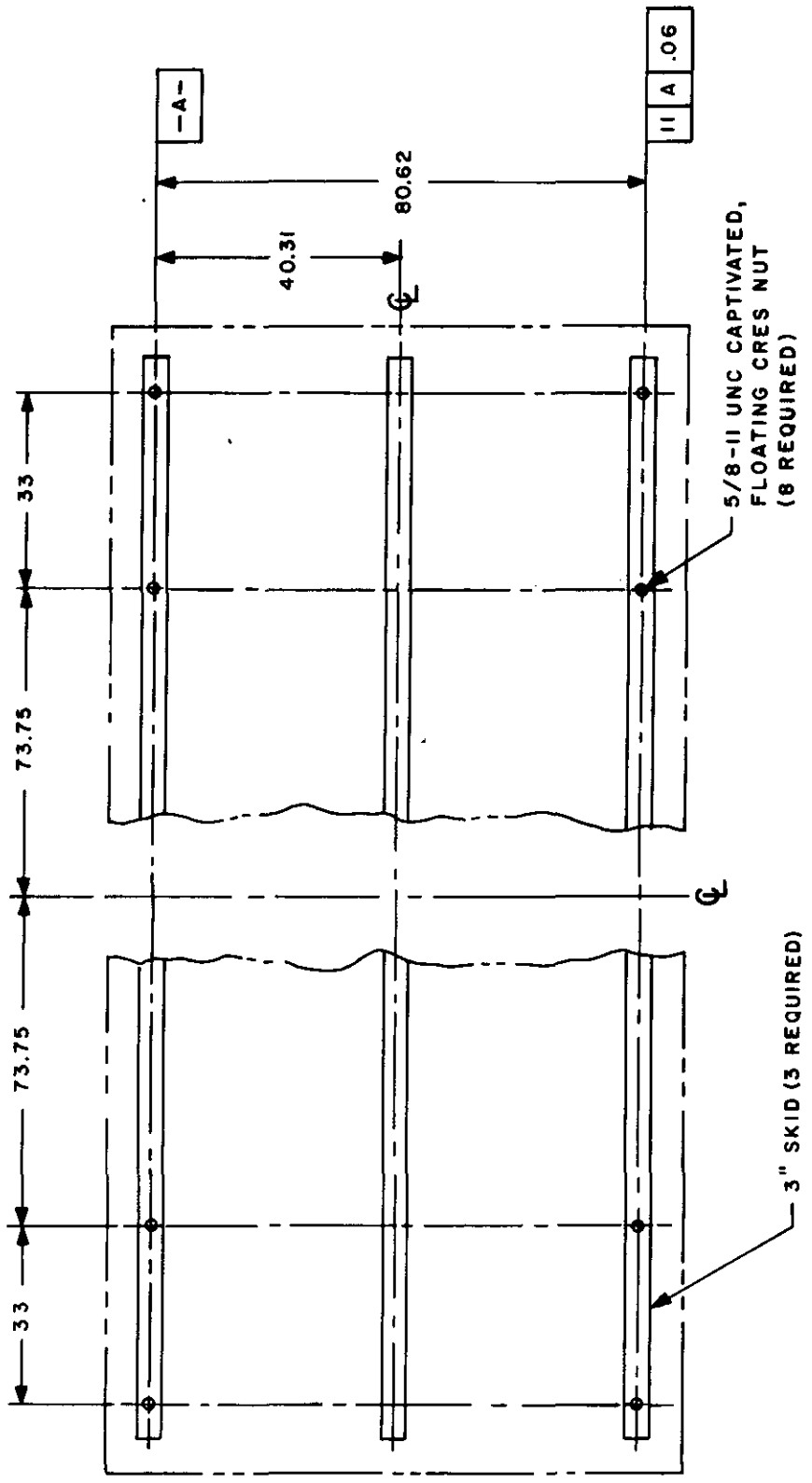


FIGURE 5. SKID LOCATION

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DOCUMENT IDENTIFIER AND TITLE
MIL-M-81957/5(AS) MBOILE FACILITY, SHELTER, INSTRUMENTATION AND CONTROL TURBO SHAFT ENGINE TEst System

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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